

Modelling of Hammer and Piano String Interaction

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Abstract—This report aims to model the interaction between the hammer and string on a piano. The model is based off of the one presented by Bensa *et al.* [1]. The model is discretized in the time domain using forward Euler, backward Euler, and trapezoidal rule (also called Crank Nicholson scheme [1]). The accuracy, stability, and CPU time of these three approaches are compared. The model parameters are then experimented with, to determine their influence on the note produced.

Index Terms—Numeric simulation, string interaction, Newton's Method, Runge-Kutta.

I. INTRODUCTION

THE sound from an acoustic piano is produced by pressing the key, which is mechanically connected to a hammer, which then strikes a string [1]. A diagram showing this mechanism is displayed in Figure 1. The vibrations of the

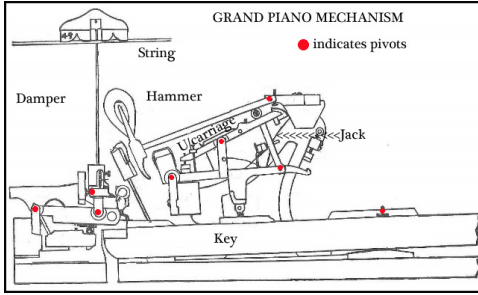


Fig. 1: Diagram showing the mechanism for the piano key, hammer, and string [2]

string in turn vibrate the surrounding air to produce the sound we hear. This hammer and string interaction appears simple at first glance, however the physics behind it are much more complex. The rise of electric pianos has motivated research into a way to model this interaction. This will allow piano sounds to be synthesized accurately instead of just playing pre-recorded notes. There has been much work in this important field to date [3]–[8]. This work has mainly been done on constructing the model and identifying key parameters for it experimentally.

This report will aim to replicated the model from Bensa, Gipoloux, and Kronland-Martinet's 2005 paper [1]. We construct the model as specified by Bensa *et al.*. We then use forward Euler, backward Euler, and trapezoidal rule to step through time. The accuracy, stability, and CPU time of these approaches are calculated and compared. The parameters of the model are then adjusted to determine their influence of the piano sound. The paper also goes one step further by attempting to optimize for the correct parameters but that is beyond the scope of this project.

II. HAMMER-STRING MODEL

Their proposed string model is as follows,

$$\frac{\partial^2 y_s}{\partial t^2} = c^2 \frac{\partial^2 y_s}{\partial x^2} - \kappa^2 \frac{\partial^4 y_s}{\partial x^4} - 2b_1 \frac{\partial y_s}{\partial t} + 2b_2 \frac{\partial^3 y_s}{\partial x^2 \partial t} \quad (1)$$

where the symbol meanings are the same as those presented in Table 1 of the paper [1]. The hammer is incorporated into the model by using a windowed forcing function. This function also models the felt compression experienced as the hammer strikes the string. The full model (with reduced order by substituting y_s for v_s) is shown below.

$$\frac{\partial y_s}{\partial t} = v_s \quad (2a)$$

$$\frac{\partial v_s}{\partial t} = c^2 \frac{\partial^2 y_s}{\partial x^2} - \kappa^2 \frac{\partial^4 y_s}{\partial x^4} - 2b_1 v_s + 2b_2 \frac{\partial^2 v_s}{\partial x^2} + g(x, x_0) f_H(y_s - y_H, v_s - v_H) \quad (2b)$$

$$\frac{\partial y_H}{\partial t} = v_H \quad (2c)$$

$$\frac{\partial v_H}{\partial t} = -\frac{1}{M_H} F_H(y_s - y_H, v_s - v_H) \quad (2d)$$

The windowing function, $g(x, x_0)$, is used to simulate the shape of the hammer interacting with the string and to smooth an otherwise rectangular hammer shape. The hammer force functions F_H and f_H are,

$$F_H(u, v) = \begin{cases} K_H(-u)^p(1 - \mu v), & \text{for } u < 0 \\ 0, & \text{for } u \geq 0 \end{cases} \quad (3a)$$

$$f_H(u, v) = F_H \left\{ \frac{M_S}{L} \int_L g(x, x_0) dx \right\}^{-1} \quad (3b)$$

which is slightly modified from as they appear in Bensa *et al.* [1]. This modification is done to make the directions of hammer movement and string position align correctly.

The boundary conditions for the string are

$$y_s(x, 0) = y_{s0}(x) \quad (4a)$$

$$v_s(x, 0) = v_{s0}(x) \quad (4b)$$

$$y_H(0) = y_s(x_0, 0) \quad (4c)$$

$$v_H(0) = v_{H0} \quad (4d)$$

$$y_s(0, t) = y_s(L, t) = 0 \quad (4e)$$

$$\frac{\partial^2 y_s}{\partial x^2}(0, t) = \frac{\partial^2 y_s}{\partial x^2}(L, t) = 0 \quad (4f)$$

TABLE I: Parameters used for C4 Note

N	100
f_s	44.1 kHz
κ	1.1466
μ	0.1998
c	323.101
b_1	0.568
b_2	1.628×10^{-6}
K_H	1.09949×10^{10}
p	2.56
M_H	0.002756

The spatial discretization is performed by a classical centered finite differences scheme [1]. This has the form

$$\frac{\partial^2 X}{\partial x^2} \approx h^{-2}(X_{i-1} - 2X_i + X_{i+1}) \quad (5a)$$

$$\frac{\partial^4 X}{\partial x^4} \approx h^{-4}(X_{i-2} - 4X_{i-1} + 6X_i - 4X_{i+1} + X_{i+2}) \quad (5b)$$

for the second and fourth spatial derivative. The boundary conditions in eq. (4) allow us to determine the fourth derivatives at the endpoints by creating virtual nodes. These nodes are non-physical but allow the fourth derivative to be determined.

III. VALIDATION OF MODEL

In order to validate our code to emulate the model we replicated Figures 2 to 4 created by Bensa *et al.* [1]. The reproduced figures are shown in Figures 2 to 4. The note produced is intended to be C4 ("middle C", at approximately 262 Hz). The parameters for the simulation are listed in Table I. They are the ones from the additional notes provided by Prof. Triverio on the project.

As we can see, the reproduced figures agree well with the paper so we can be confident that the model is correct. We are able to see that with increased initial velocity, the hammer exerts more force on the string and appears to bounce on the string while in contact with it. We can also see that the advanced hammer felt model used by Bensa *et al.* allows for more pronounced hysteresis of the felt with higher initial velocities.

IV. COMPARISON OF SOLUTION METHODS

A comparison of the CPU time, stability and accuracy between different solution methods — forward Euler (FE), backward (BE) Euler and trapezoid rule (TR) — is investigated.

For accuracy comparisons, the TR solution with a high number of spatial discretizations (i.e. $N = 500$ as opposed to the original 100) is taken as a stand-in for the analytic solution (which is not readily available). To evaluate the accuracy, FE output (string displacement at a certain point) was compared with TR output for 5ms. The percent error between the two would give a good idea how accurate the solution would be. A similar idea was employed for BE and TR. Numeric cost is determined by comparing elapsed CPU time.

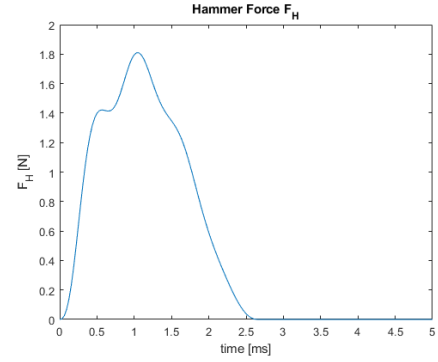
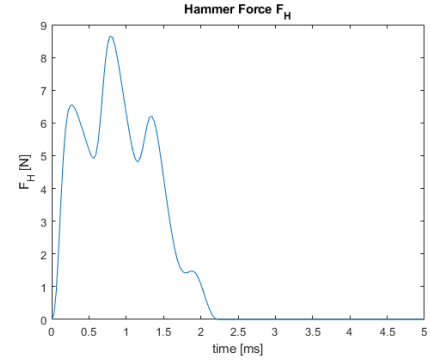
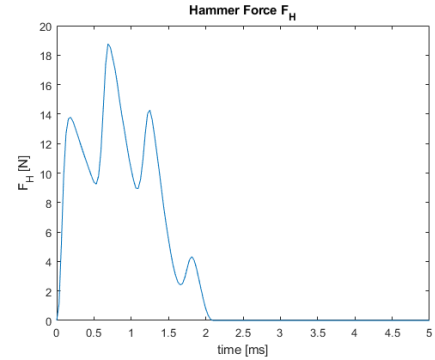
(a) $v_H = 0.5m/s$ (b) $v_H = 2m/s$ (c) $v_H = 4m/s$

Fig. 2: Hammer force curves for different initial velocities. Reproduction of Fig 2. in Bensa *et al.* [1].

A. Forward Euler (FE)

The FE method is a first-order, explicit method of the Runge-Kutta class that is simple and easy to implement. Each time step of FE is computed solely by a matrix vector product, making it fast. However, its accuracy and stability characteristics are known to be poor, which limits its usefulness.

To evaluate the stability of FE, the sampling frequency was varied in multiples of the sampling frequency 44.1kHz. The stability of the method was reflected in the amount of time it took for the system to become unstable due to accumulated errors. This was determined by the string having oscillation amplitudes over 10m. A plot of the results is shown in Figure 5. There appears to be a linear dependence on the amount of simulatable time and the sampling frequency.

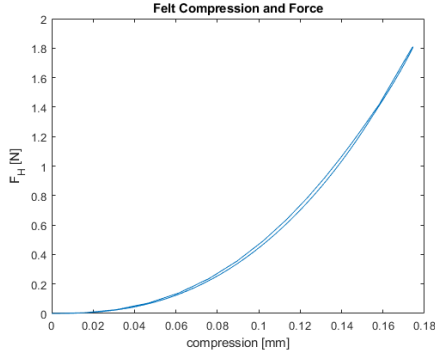
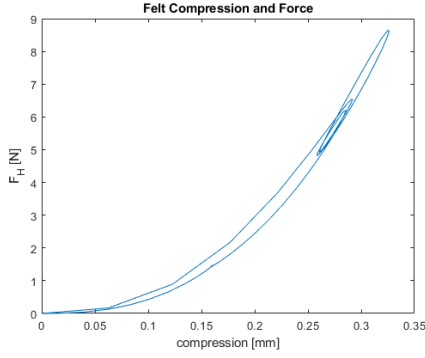
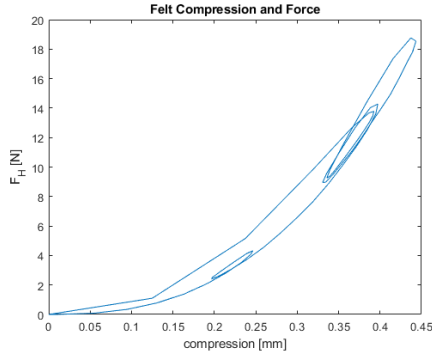
(a) $v_H = 0.5 \text{ m/s}$ (b) $v_H = 2 \text{ m/s}$ (c) $v_H = 4 \text{ m/s}$

Fig. 3: Felt Compression and hysteresis. Reproduction of Fig 3. in Bensa et al. [1].

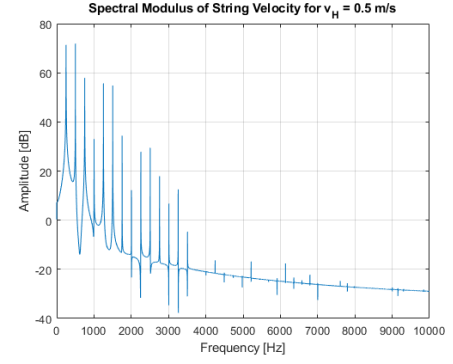
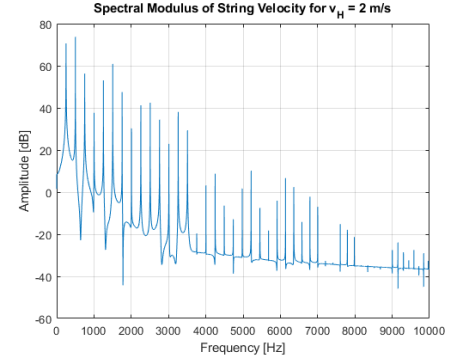
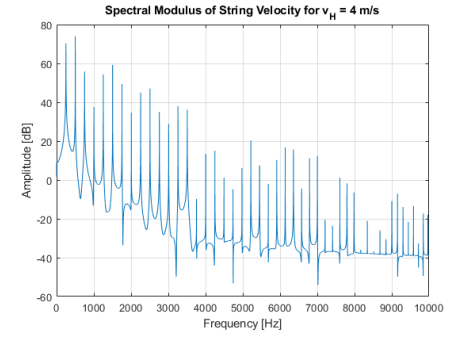
(a) $v_H = 0.5 \text{ m/s}$ (b) $v_H = 2 \text{ m/s}$ (c) $v_H = 4 \text{ m/s}$

Fig. 4: Spectra of the velocity signal for different hammer velocities. Reproduction of Fig 4. in Bensa et al. [1].

To evaluate the accuracy and CPU time for FE, the simulation of 5ms was run at a sampling frequency of $500 \times 44.1 \text{ kHz}$ to ensure stability and then the output was just sampled every 500 steps. The results for different spatial discretizations is presented in Figure 6. The accuracy of the algorithm is computed by measuring the percent difference in the output (i.e. the position of one of the string points over time) between the result with the finest TR discretization and the discretization in question. There seems to be an overall trend in increasing accuracy with N . Because the simulation was 5ms there were many errors accumulated. This results in the large error at roughly 86 percent. The CPU time unsurprisingly increases with larger N . It seems to taper off for larger N . At $N=100$ There is a decrease in CPU time.

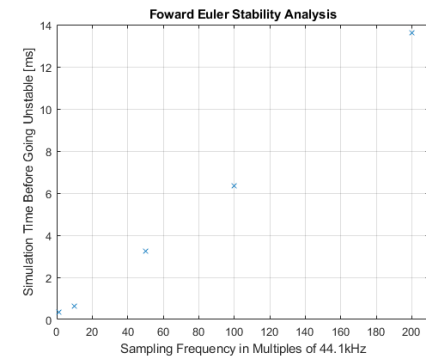
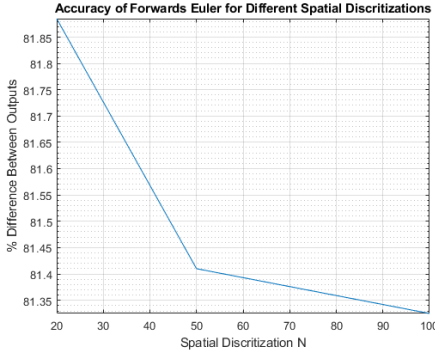
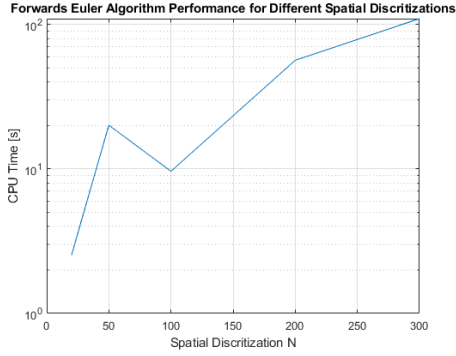


Fig. 5: Stability analysis for FE

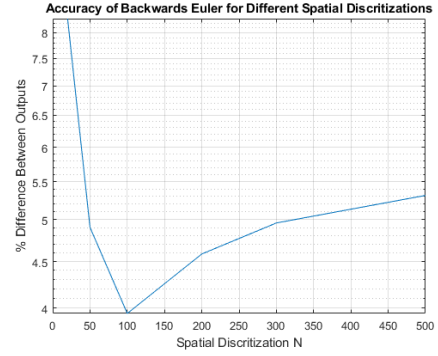


(a) FE Accuracy

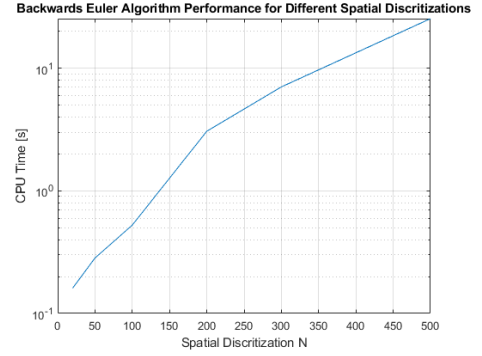


(b) FE CPU Time

Fig. 6: Accuracy and CPU Time for BE



(a) BE Accuracy



(b) BE CPU Time

Fig. 7: Accuracy and CPU Time for BE

B. Backward Euler (BE)

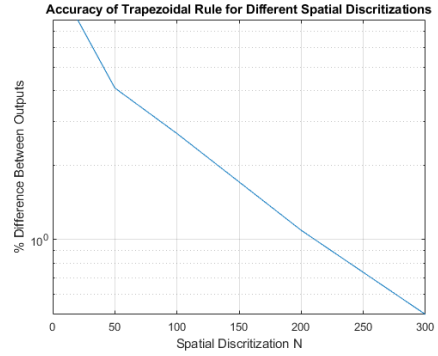
The BE method is an implicit method resulting from a modification of FE, leading to improved stability characteristics. Its accuracy as with FE remains first-order. The method is by its nature always stable if the system is stable (this is true in our case).

The accuracy and CPU time for BE for different spatial discretizations is presented in Figure 7. The accuracy of the algorithm is computed by measuring the percent difference in the output (i.e. the position of one of the string points over time) between the result with the finest TR discretization and the discretization in question. There seems to be an overall trend in increasing accuracy with N except for at $N=100$ where there is the best accuracy. The CPU time unsurprisingly increases with larger N . It seems to taper off for larger N .

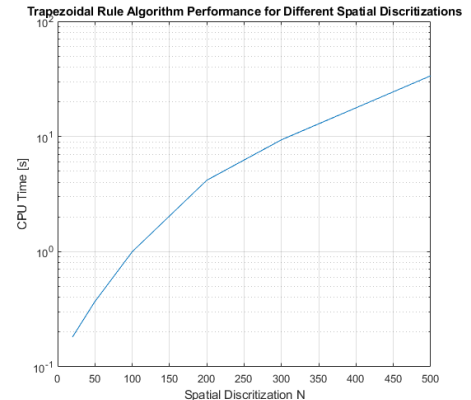
C. Trapezoid Rule (TR)

This method is implemented in the reference document [1]. The TR is expected to be the most accurate of all three methods presented with good stability characteristics and second-order accuracy. For all of the comparisons and parameter variations discussed from this section onwards, the TR method is used. The method is by its nature always stable if the system is stable (this is true in our case).

The accuracy and CPU time for TR for different spatial discretizations is presented in Figure 8. The accuracy of the algorithm is computed by measuring the percent difference in the output (i.e. the position of one of the string points



(a) TR Accuracy



(b) TR CPU Time

Fig. 8: Accuracy and CPU Time for TR

over time) between the result with the finest discretization and the discretization in question. As we can see, increasing the sampling drastically reduces the accuracy of the solution. That being said, the solution for fairly coarse discretization at $N=20$ is still quite close with only about 7.77 percent difference.

The CPU time does increase quite rapidly for increasing N . However, we can see that it begins to taper off when N is very large (roughly 500).

D. Conclusions on Accuracy, CPU time, and Stability

Unsurprisingly, TR is the most accurate followed by BE and then FE. However, BE takes less time to execute than TR. It is impossible to compare FE and the two implicit methods because the sampling frequency had to be increased so much. Despite this we can say that the computational time for the equivalent tasks was less accurate, and took longer or FE. FE was found to have a linear dependence on stability with respect to N .

Interestingly, near $N=100$ we see an increase in accuracy for BE and a decrease in CPU time for FE. We are not sure of the cause of this. We can speculate that perhaps there is some speed ups MATLAB is able to make with systems of this size.

V. EFFECTS OF MODIFYING PARAMETERS

In this section various parameters are adjusted to determine their effect. A change in simulation parameters is considered to have "no effect" if no difference can be discerned in the resulting sound file (although this is subjective depending on the listener, and somewhat complicated by the coarseness of the audio conversion). The results are plotted for felt compression, hammer force, frequency components, spectral modulus of velocity and final string displacement.

Total simulation time was reduced to 2.5 ms from 5 ms for convenience (the impact on final result was not noted to be qualitatively different); average CPU time to compute a note was approximately 20 s. The starting parameters for each scenario considered are the same as those used in [1] and Table I, with hammer velocity $v_H = 2.0m/s$, $N = 100$ spatial discretizations, $dt = 1/44100$ s — this result is taken to be the "original" solution. This tone is also used as the reference tone for determining if any audible change has arisen.

Graphs relating to observations are provided in the appendix.

A. Fineness of Spatial Discretization N

The fineness of spatial discretization is experimented with. Scenarios of $N = 20, 40, 60, 80, 100, 200$ are considered. As the number of discretizations increases, the greater spatial resolution leads to an increased brightness of sound; for a coarse discretization, the sound is more coarse and noticeably dimmer in pitch. This is likely due to a lower number of discretizations being unable to accommodate higher frequency vibrations.

The effect on accuracy of the solution is also investigated. Deviations in accuracy of the resultant string position and hysteresis curve are observed, although the effect on the resulting sound is not pronounced.

B. Fineness of Timestep dt

The fineness of time discretization is experimented with. The original value of dt used corresponds to $1/f_{max}$ where $f_{max} = 44100$ Hz, a nominal upper limit of human hearing. New timesteps of 0.5, 0.75, 1.25, 1.5, 1.75 and 2 times dt are considered. The effect of a longer timestep is an increase in pitch, evident from analyzing the frequency components of the resulting tone.

For timesteps beyond $2dt$, Newton's method for the non-linear component of the hammer-string interaction fails to converge.

dt_{factor} is multiplicative factor by which dt was altered from the value used to simulate C4.

C. String Stiffness Coefficient κ

The string stiffness coefficient κ characterizes the string's resistance to an applied force, ie. hammer strike. It is associated with the 4th-order spatial derivative term in (1), which arises from the PDE describing a vibrating bar ($u_{tt} - u_{xxxx} = 0$). Therefore, adjusting κ also allows for the effect of the 4th-order term to be determined.

Increasing the spring stiffness coefficient leads to a more "hollow", metallic quality to the sound (as could be expected, since the material is more rigid). The effect of adjusting κ within 0 to 2 times its original value is however difficult to perceive by ear; the effect is more produced for much greater changes (ie. increase by a factor of 4 or 10).

κ_{factor} is multiplicative factor by which dt was altered from the value used to simulate C4.

D. Felt Stiffness Coefficient K_H

The force due to the compression of felt can be modelled as a type of nonlinear string interaction of the form $f_H = K_H x^p (1 + \mu \frac{dx}{dt})$, where K_H is the felt stiffness coefficient and p is the felt stiffness exponent. Increasing K_H and p has the effect of increasing the overall stiffness of the felt material.

Increasing the stiffness of the felt material has a similar effect to increasing the string stiffness coefficient, leading to a more "hollow" and metallic sound quality. The effect of increased stiffness is readily visible for the hysteresis curve, where felt compression is reduced.

$K_{H,factor}$ is multiplicative factor by which dt was altered from the value used to simulate C4.

E. Felt Loss Coefficient μ

The felt loss coefficient μ in the string interaction $f_H = K_H x^p (1 + \mu \frac{dx}{dt})$ characterizes the additional compression forces felt due to the speed at which the felt is impacted. Increasing the felt loss coefficient has the effect of producing a "flatter" sound than the original note.

μ_{factor} is multiplicative factor by which μ was altered from the value used to simulate C4.

F. String Loss Parameters b_1 , b_2

The string loss parameters b_1 and b_2 are associated with frequency dependent losses. Adjusting them allows the effect of the frequency dependent components in (1) to be determined.

Loss term b_1 is associated with the time derivative dy_s/dt , leading to greater loss for higher frequency oscillations. The effect of increasing b_1 is a shorter, more staccato note. The effect of increasing b_2 is similar, although b_2 must be increased by approximately 5 orders of magnitude before a significant change is audible. This suggests that the relative contribution of b_1 (associated with the 1st order time derivative) to the end result is significantly greater (in comparison to the 3rd order derivative).

$b_{1,factor}$ is multiplicative factor by which b_1 was altered from the value used to simulate C4 (and similarly with b_2).

G. Change in Hammer Velocity

Changing the hammer velocity (i.e. pressing the key harder) increases the amplitude of high frequency harmonics. This is evident when we examine the differences between hammer velocities in Figure 4. Having an initial velocity of 4 m/s produces higher amplitude harmonics when compared to a velocity of 0.5 m/s. When playing back the sound we can observe a louder and brighter note; the brightness is due to the presence of these harmonics ("modification of the spectral shape"). This can be easily verified by experimenting with a physical piano.

H. Plucked String Condition

The "plucking" motion of the piano string can be approximated by setting the initial displacement to a triangle function, with the "peak" representing the location at which the string is being pulled.

The effect of the plucking initial condition is a more "discrete", discontinuous set of tones reminiscent of an actual plucked string; this likely arises from non-smooth nature of the triangle function compared to the piano hammer hit.

VI. CONCLUSION

In this report we were able to create a functioning hammer and piano string model. The time dimension was then discretized using TR, FE, and BE. The accuracy, CPU time, and stability of these methods were compared and TR seems to be the best compromise between the three parameters.

Experimenting with various simulation parameters allowed for the contributions of each component of the wave equation to be identified, which allows for a wide range of tones to be created using this model. The major barrier to accurately replicating notes is however the determination of system parameters, which can only be done experimentally and with difficulty. Nevertheless, as a result of the ongoing research into modelling pianos the public can expect high performing electric pianos in the future.

ACKNOWLEDGMENT

The authors would like to thank Prof. Piero Triverio for his helpful instruction and guidance which made this project possible.

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APPENDIX A PARAMETER VARIATION RESULTS

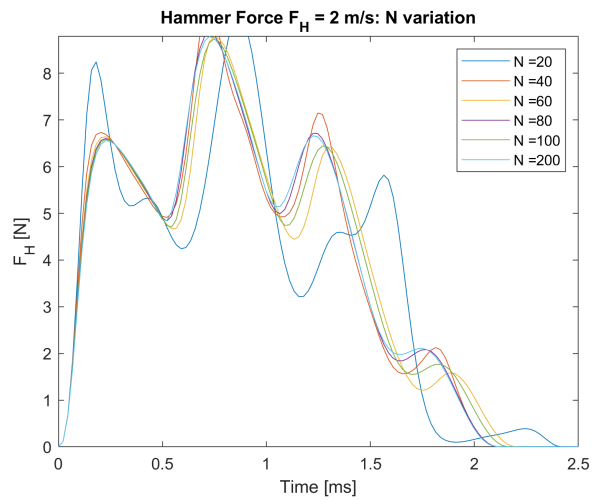


Fig. 9: Results for N discretizations variation (and following 4 figures)

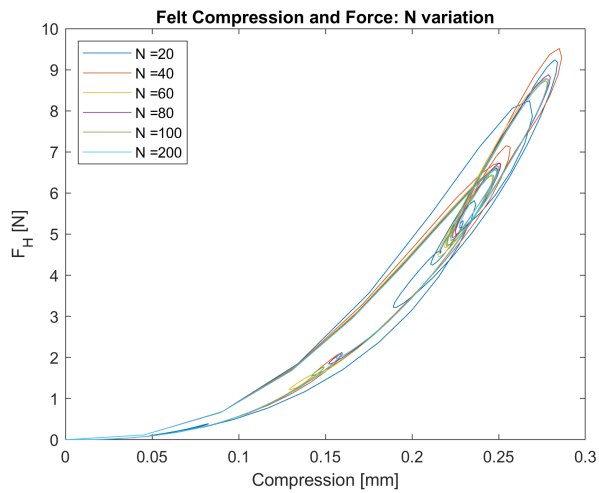


Fig. 10

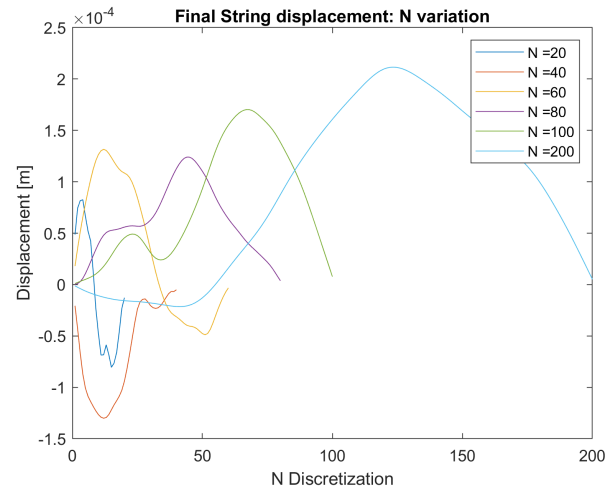


Fig. 11

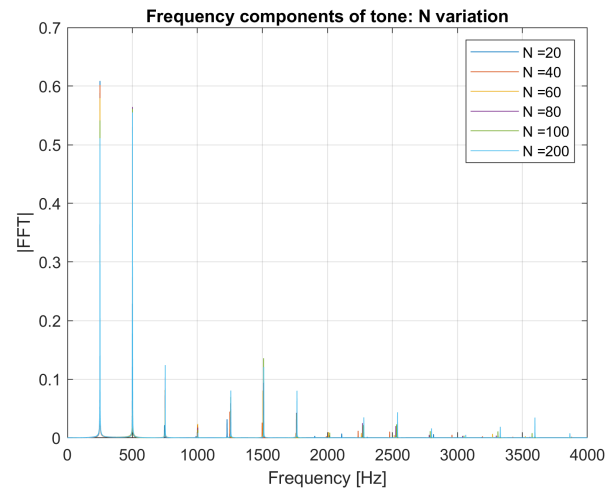


Fig. 12

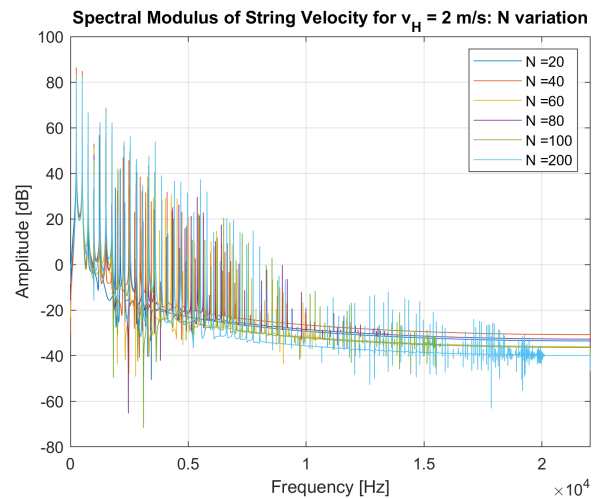


Fig. 13

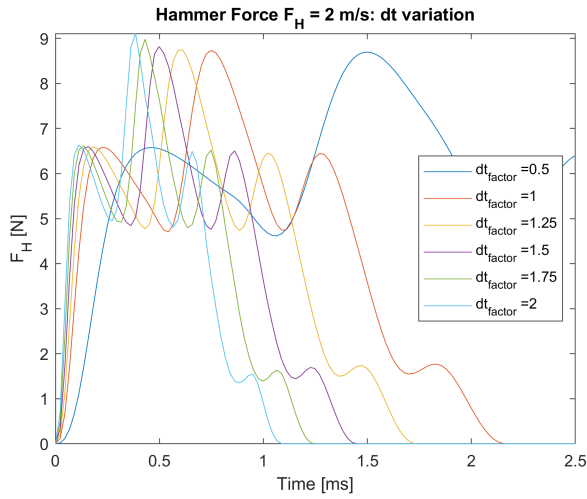


Fig. 14: Results for dt timestep variation (and following 4 figures)

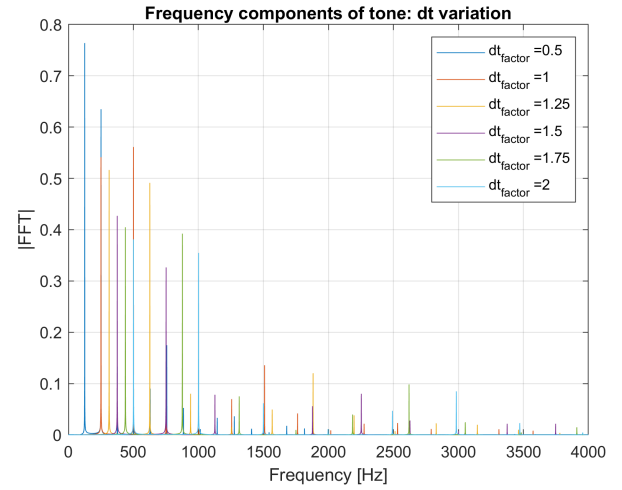


Fig. 17

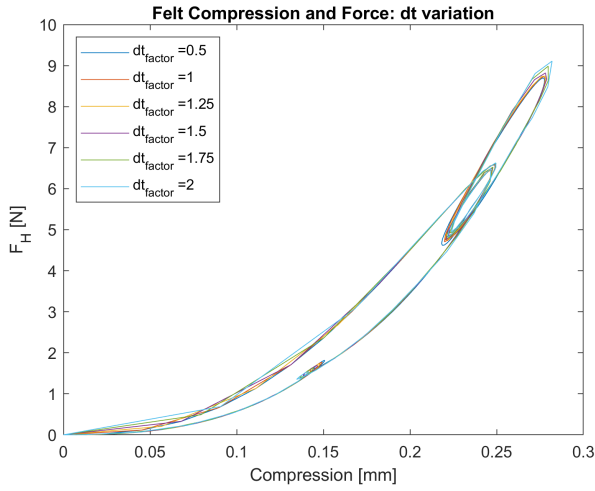


Fig. 15

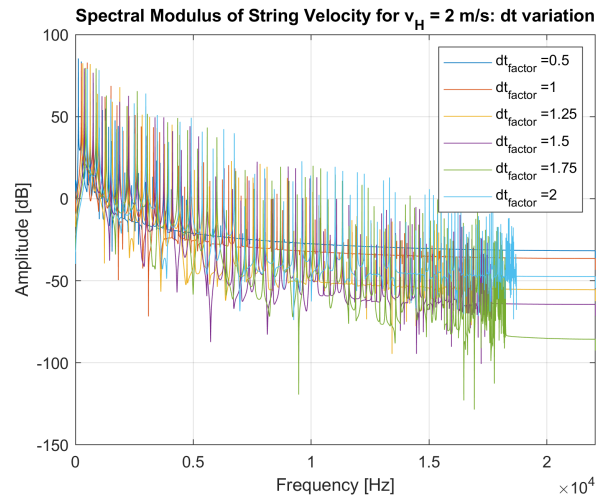


Fig. 18

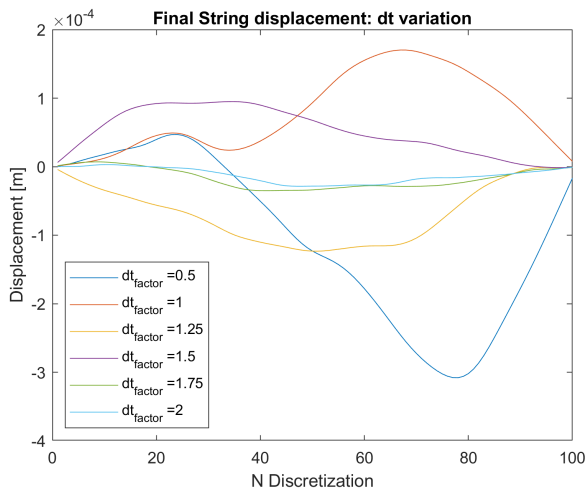


Fig. 16

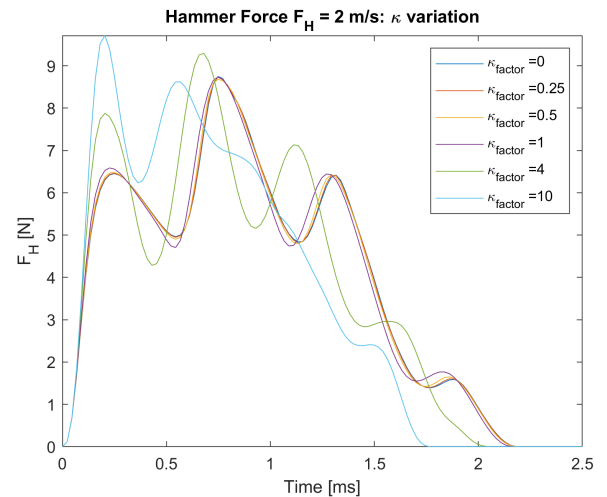


Fig. 19: Results for κ string stiffness variation (and following 4 figures)

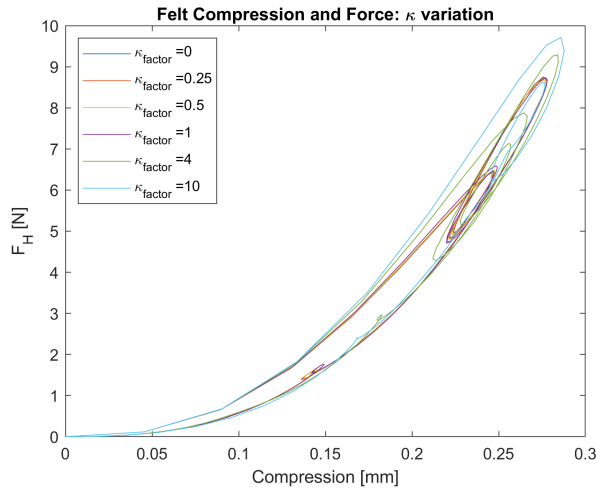


Fig. 20

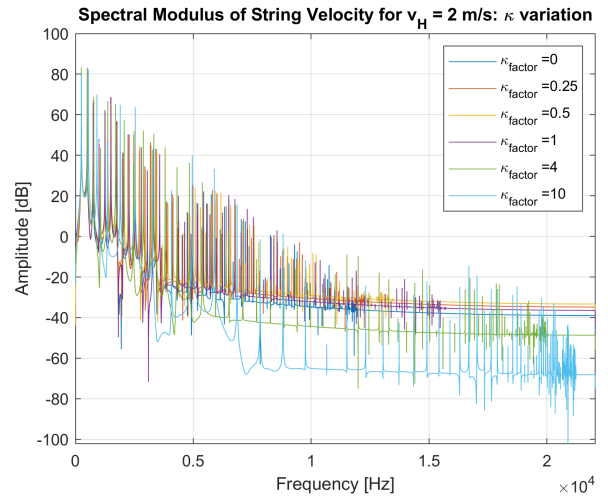


Fig. 23

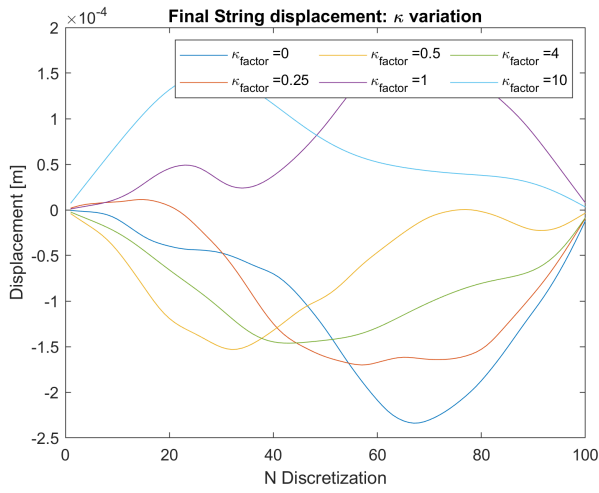


Fig. 21

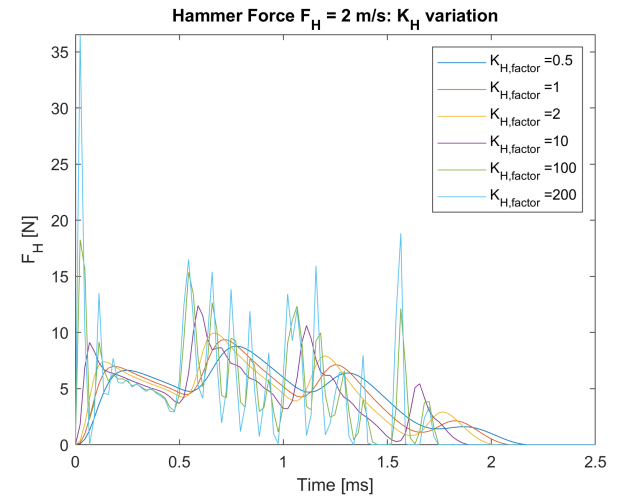
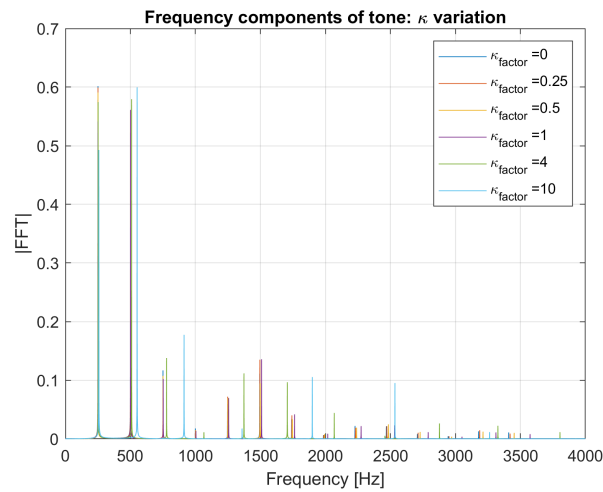
Fig. 24: Results for K_H felt stiffness variation (and following 4 figures)

Fig. 22

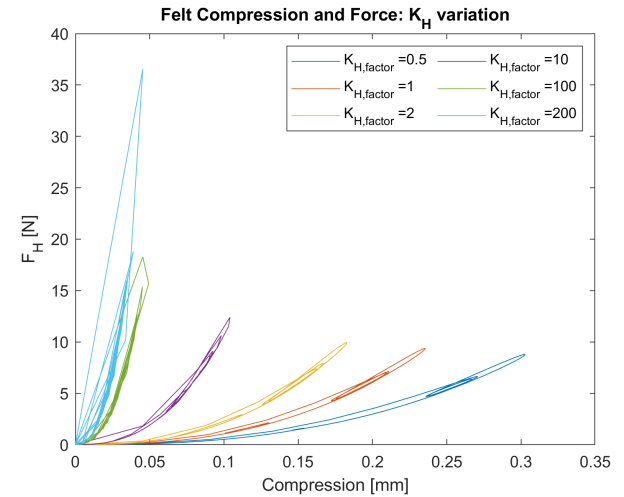


Fig. 25

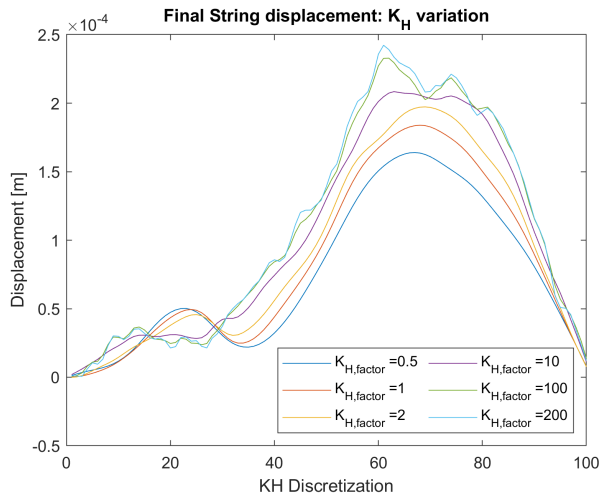


Fig. 26

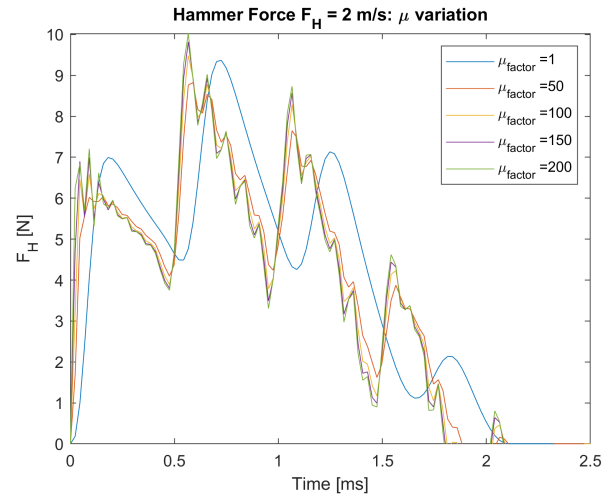
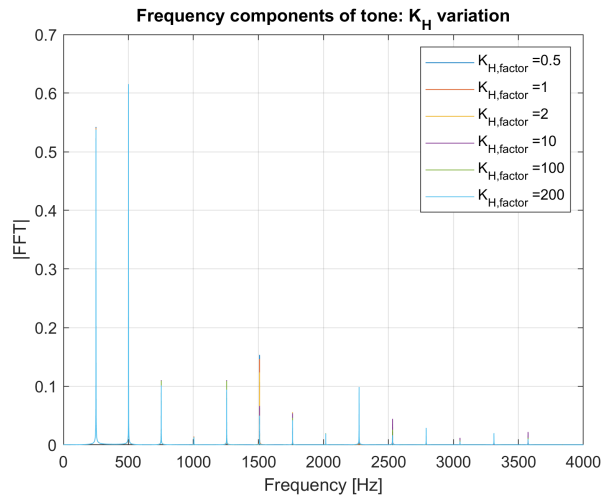
Fig. 29: Results for μ felt loss coefficient variation (and following 4 figures)

Fig. 27

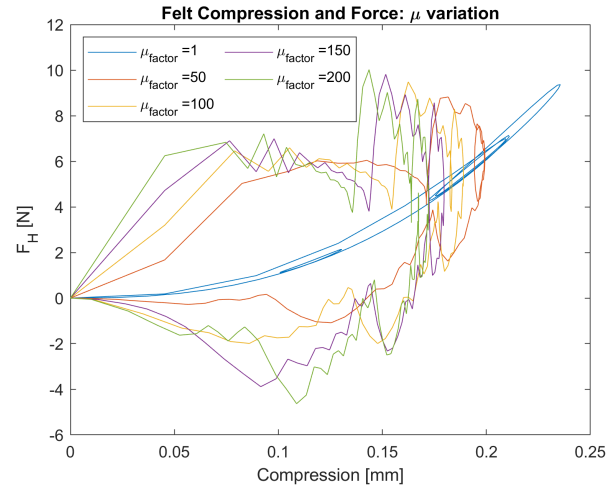


Fig. 30

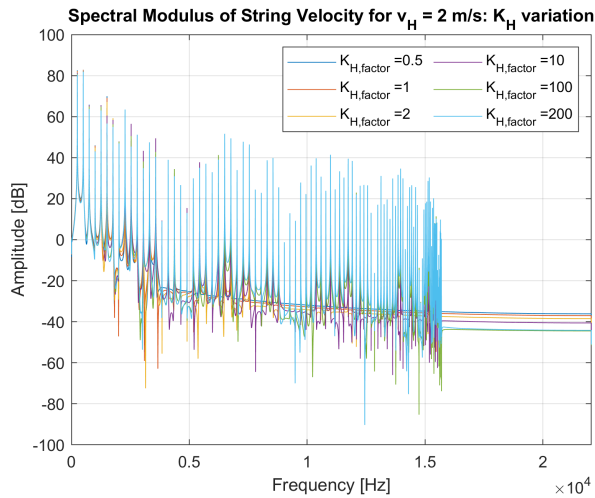


Fig. 28

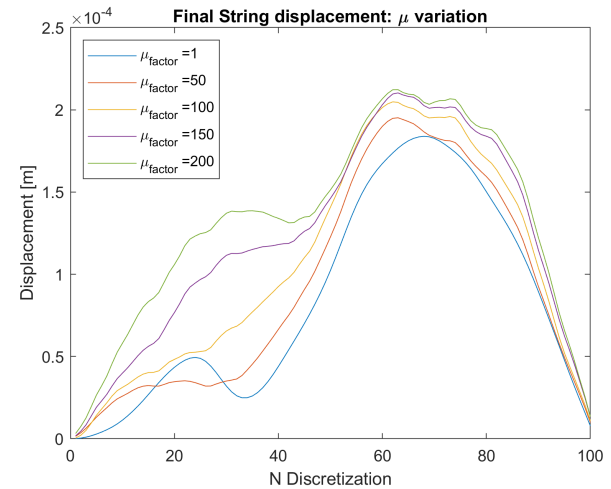


Fig. 31

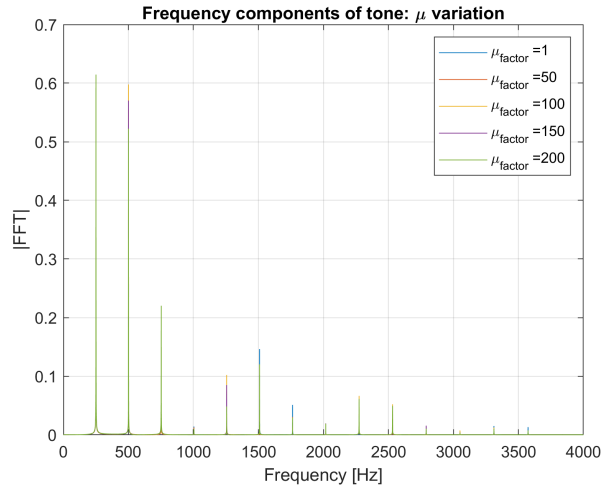


Fig. 32

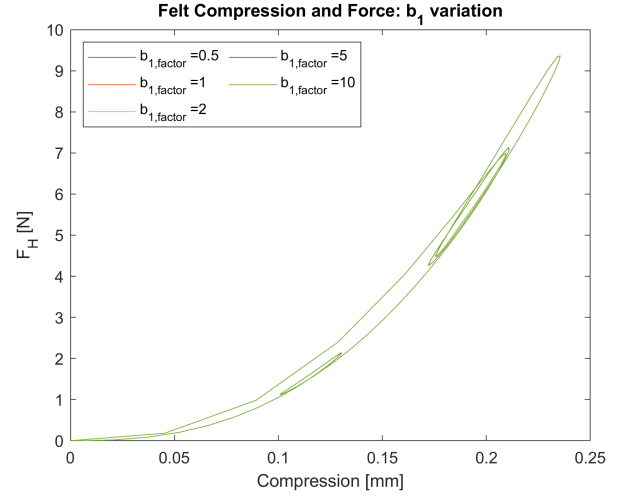


Fig. 35

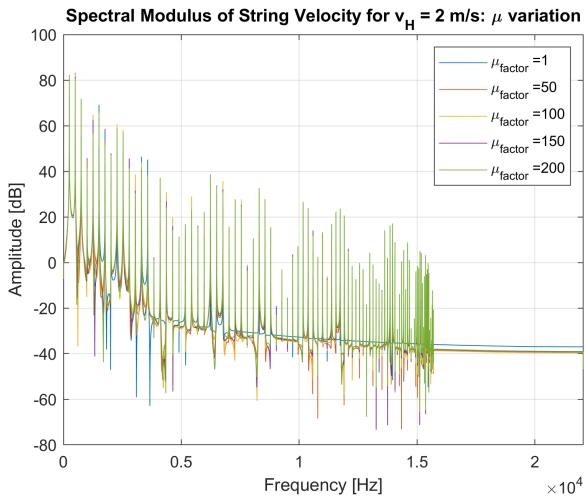


Fig. 33

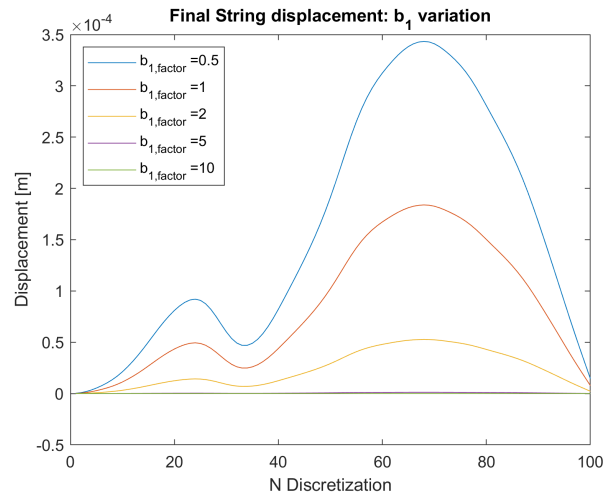


Fig. 36

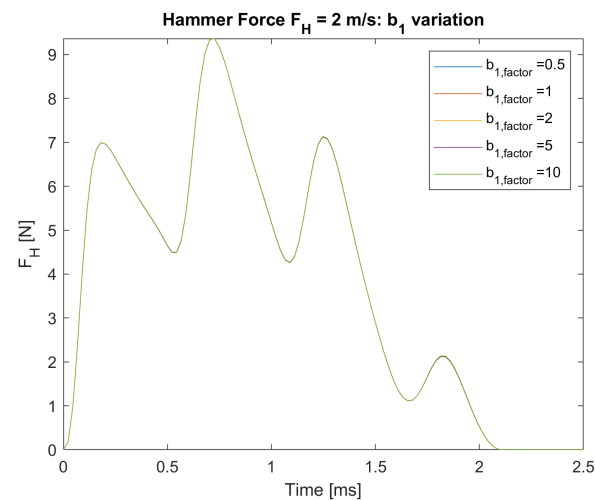
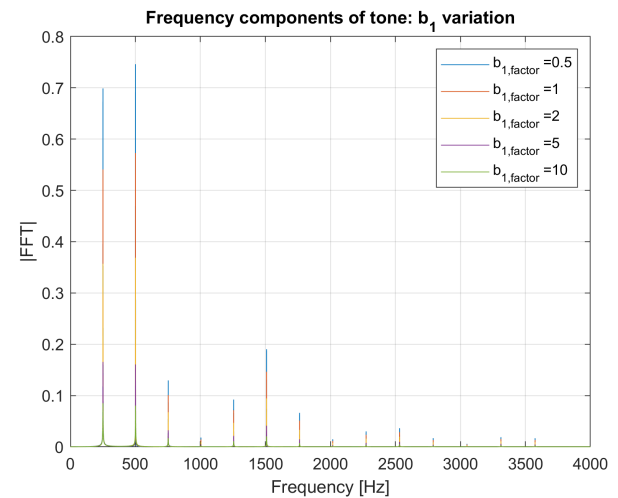
Fig. 34: Results for b_1 string loss coefficient variation (and following 4 figures)

Fig. 37

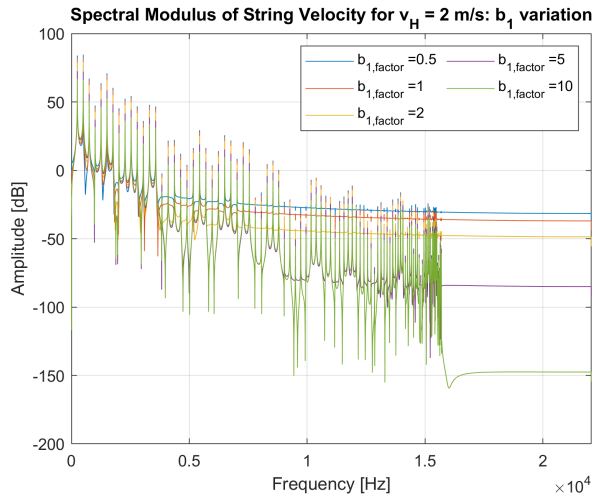


Fig. 38

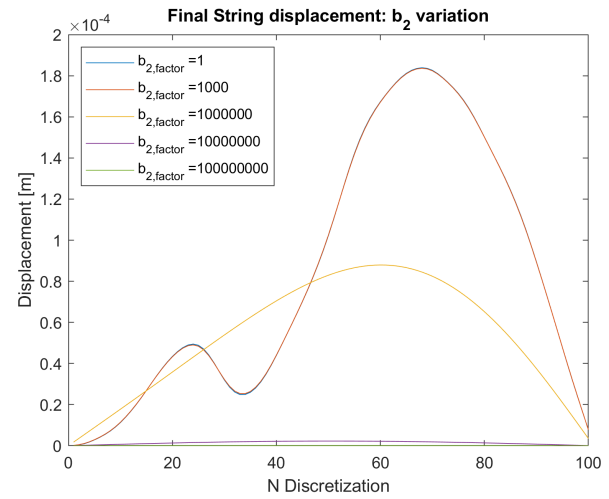


Fig. 41

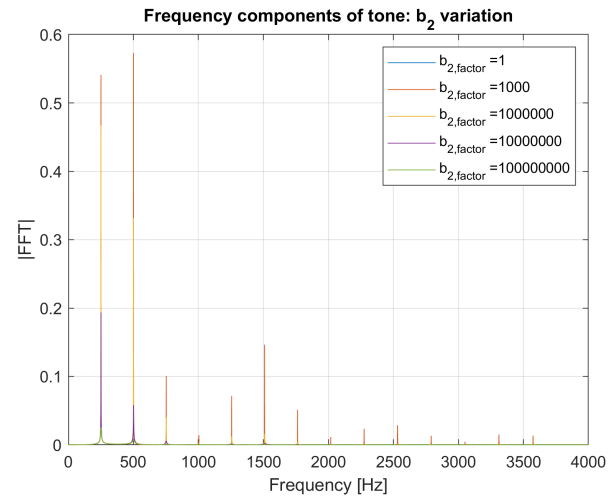
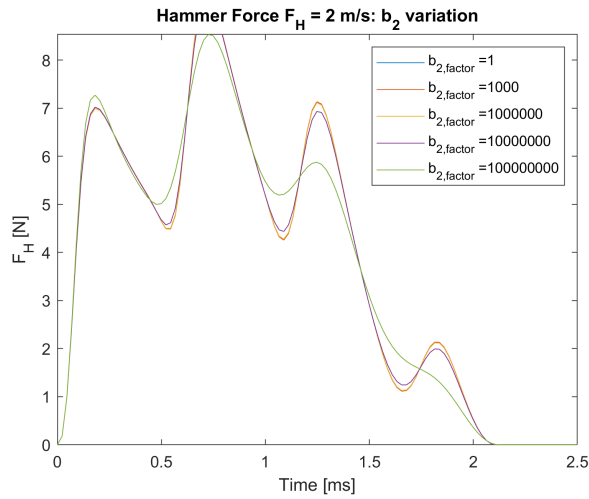


Fig. 42

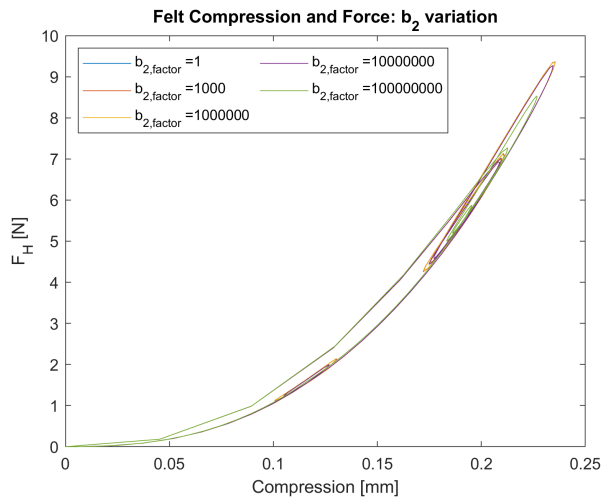
Fig. 39: Results for b_2 string loss coefficient variation (and following 4 figures)

Fig. 40

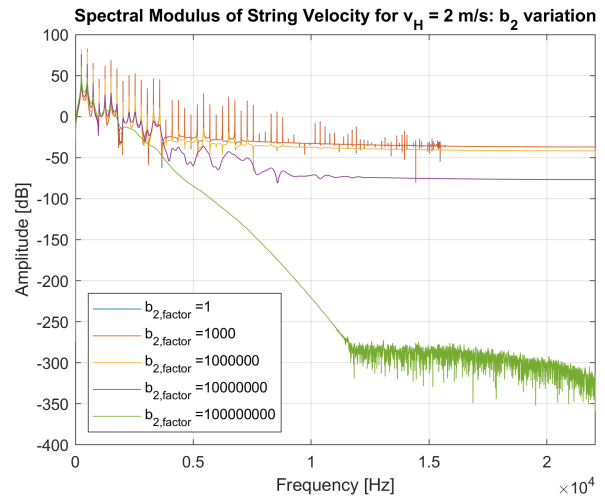


Fig. 43